

MCCHORD AFB, WA, USA

WMO: 742060

Lat: 47.150N

Lon: 122.483W

Elev: 322

StdP: 14.53

Time Zone: -8.00 (NAP)

Period: 94-19

WBAN: 24207

Annual Heating, Humidification, and Ventilation Design Conditions

Coldest Month	Heating DB		Humidification DP/MCDB and HR						Coldest Month WS/MCDB				MCWS/PCWD to 99.6% DB		WSF
			99.6%			99%			0.4%		1%				
	99.6%	99%	DP	HR	MCDB	DP	HR	MCDB	WS	MCDB	WS	MCDB	MCWS	PCWD	
(a)	(b)	(c)	(d)	(e)	(f)	(g)	(h)	(i)	(j)	(k)	(l)	(m)	(n)	(o)	(p)
12	21.0	24.8	13.3	11.0	27.2	19.0	14.5	27.8	25.6	47.0	22.6	45.7	2.5	180	0.531

Annual Cooling, Dehumidification, and Enthalpy Design Conditions

Hottest Month	Hottest Month DB Range	Cooling DB/MCWB						Evaporation WB/MCDB						MCWS/PCWD to 0.4% DB	
		0.4%		1%		2%		0.4%		1%		2%			
		DB	MCWB	DB	MCWB	DB	MCWB	WB	MCDB	WB	MCDB	WB	MCDB	MCWS	PCWD
(a)	(b)	(c)	(d)	(e)	(f)	(g)	(h)	(i)	(j)	(k)	(l)	(m)	(n)	(o)	(p)
7	23.2	86.9	65.7	83.0	64.4	79.5	62.9	67.7	82.1	66.0	79.4	64.3	76.6	8.1	30

Dehumidification DP/MCDB and HR									Enthalpy/MCDB						Extreme Max WB
0.4%			1%			2%			0.4%		1%		2%		
DP	HR	MCDB	DP	HR	MCDB	DP	HR	MCDB	Enth	MCDB	Enth	MCDB	Enth	MCDB	
(a)	(b)	(c)	(d)	(e)	(f)	(g)	(h)	(i)	(j)	(k)	(l)	(m)	(n)	(o)	(p)
62.9	86.9	70.5	61.2	81.6	69.0	59.6	77.0	67.5	32.1	82.3	30.7	79.3	29.5	76.7	73.0

Extreme Annual Design Conditions

Extreme Annual WS			Extreme Annual Temperature				n-Year Return Period Values of Extreme Temperature							
1%	2.5%	5%	Mean		Standard Deviation		n=5 years		n=10 years		n=20 years		n=50 years	
			Min	Max	Min	Max	Min	Max	Min	Max	Min	Max	Min	Max
(a)	(b)	(c)	(e)	(f)	(g)	(h)	(i)	(j)	(k)	(l)	(m)	(n)	(o)	(p)
20.2	17.6	15.5	13.8	93.5	5.0	3.9	10.2	96.2	7.2	98.5	4.4	100.7	0.8	103.5
			13.2	69.8	5.0	2.0	9.6	71.2	6.7	72.4	3.9	73.5	0.3	75.0
			DB											
			WB											

Monthly Climatic Design Conditions

		Annual	Jan	Feb	Mar	Apr	May	Jun	Jul	Aug	Sep	Oct	Nov	Dec	
		(d)	(e)	(f)	(g)	(h)	(i)	(j)	(k)	(l)	(m)	(n)	(o)	(p)	
(6) Temperatures, Degree-Days and Degree-Hours	DBAvg	51.3	40.2	41.2	44.9	48.9	55.1	60.0	65.2	65.3	59.9	50.9	44.0	39.4	(6)
	DBStd	10.60	6.30	5.55	4.80	4.83	5.33	4.96	4.59	4.11	4.53	4.87	6.26	6.21	(7)
	HDD50	1387	305	247	170	77	13	0	0	0	1	48	195	331	(8)
	HDD65	5137	768	665	622	484	309	167	52	47	162	437	630	794	(9)
	CDD50	1866	3	2	13	43	171	299	471	474	297	75	15	3	(10)
	CDD65	139	0	0	0	0	1	16	58	56	8	0	0	0	(11)
	CDH74	2351	0	0	1	21	102	315	895	786	228	3	0	0	(12)
	CDH80	701	0	0	0	3	17	102	304	232	43	0	0	0	(13)
(14) Wind	WSAvg	6.1	5.7	6.1	7.0	6.8	6.4	6.6	6.0	5.6	5.4	5.6	6.1	6.0	(14)
(15) Precipitation	PrecAvg	41.8	5.9	4.0	4.5	3.4	2.4	1.9	0.7	1.0	1.7	4.0	6.5	5.8	(15)
	PrecMax	56.8	12.3	8.6	9.0	7.4	4.3	3.9	1.9	3.7	7.8	9.6	14.4	11.5	(16)
	PrecMin	31.2	2.4	0.2	1.5	1.0	0.3	0.3	0.0	0.0	0.1	0.7	1.7	1.2	(17)
	PrecStd	6.7	2.3	2.0	1.7	1.6	1.1	0.9	0.5	0.9	1.6	2.0	3.1	2.6	(18)
(19) Monthly Design Dry Bulb and Mean Coincident Wet Bulb Temperatures	0.4%	DB	57.8	59.3	68.5	76.9	82.2	88.9	92.8	90.9	84.9	72.6	62.9	57.3	(19)
		MCWB	52.8	49.9	54.3	59.0	62.2	65.4	67.4	67.0	63.8	58.1	56.7	54.0	(20)
	2%	DB	54.5	55.4	62.0	68.6	76.8	82.2	87.5	85.9	80.4	66.7	58.6	53.4	(21)
		MCWB	51.3	49.4	52.0	55.0	60.3	63.2	65.9	66.3	62.5	57.1	54.7	50.6	(22)
	5%	DB	52.0	53.1	57.5	63.6	72.1	76.9	82.7	82.0	76.0	63.4	55.5	50.4	(23)
		MCWB	49.4	48.5	50.1	52.4	58.1	61.1	64.8	64.4	61.5	55.9	52.4	48.4	(24)
	10%	DB	49.7	50.4	54.5	59.3	67.6	72.3	78.9	78.5	72.2	60.5	53.8	48.2	(25)
		MCWB	47.6	46.8	48.4	50.3	56.5	59.6	63.1	63.3	60.0	54.8	51.1	46.3	(26)
(27) Monthly Design Wet Bulb and Mean Coincident Dry Bulb Temperatures	0.4%	WB	55.5	54.6	56.6	60.0	64.9	68.5	70.4	69.6	67.7	62.5	58.8	54.9	(27)
		MCDB	57.5	56.5	64.2	72.8	76.7	83.9	88.6	84.5	77.2	66.7	61.0	57.0	(28)
	2%	WB	52.3	51.7	53.8	56.4	61.9	64.5	67.4	67.6	64.7	59.6	55.6	51.2	(29)
		MCDB	53.4	53.7	59.5	65.7	73.3	77.9	83.3	81.7	74.6	63.6	57.7	52.5	(30)
	5%	WB	49.8	49.2	51.5	53.9	59.5	62.2	65.8	65.8	62.8	57.5	53.4	48.7	(31)
		MCDB	51.0	51.2	55.8	60.8	69.6	74.5	80.5	79.1	72.5	61.2	55.2	50.1	(32)
	10%	WB	47.5	47.1	49.7	51.9	57.3	60.2	64.1	64.2	61.4	55.5	51.3	46.3	(33)
		MCDB	49.0	49.8	53.7	57.7	66.1	70.8	77.2	76.1	70.0	59.2	52.9	47.5	(34)
(35) Mean Daily Temperature Range	5% DB	MDBR	12.8	15.6	17.4	18.9	19.7	20.3	23.2	23.4	22.2	18.0	14.9	12.1	(35)
		MCDBR	15.1	19.9	23.9	28.0	30.4	30.9	32.8	32.0	31.3	23.6	16.6	13.3	(36)
	5% WB	MCWBR	11.4	13.5	14.7	15.4	15.4	14.0	13.8	14.0	15.7	14.7	12.1	10.6	(37)
		MCWBR	12.6	14.7	20.1	23.2	25.7	26.6	28.7	27.9	25.4	18.6	13.8	11.5	(38)
(40) Clear-Sky Solar Irradiance	taub		0.325	0.316	0.327	0.356	0.376	0.366	0.350	0.356	0.344	0.329	0.323	0.332	(40)
		taud	2.478	2.536	2.525	2.416	2.381	2.433	2.498	2.487	2.517	2.561	2.519	2.391	(41)
	Ebn at Noon		242	273	286	285	282	285	288	282	276	264	241	220	(42)
		Edh at Noon	21	24	28	34	37	35	32	32	28	23	20	20	(43)
(44) All-Sky Solar Radiation	RadAvg		325	597	887	1300	1645	1801	2012	1722	1265	718	382	268	(44)
	RadStd		49	82	108	123	138	183	153	121	117	80	50	39	(45)

Historical Trends

		Heating		Cooling			Degree-Days					
		99% DB	99% DP	1% DB	1% WB	1% DP	HDD50	HDD65	CDD50	CDD65		
		(d)	(e)	(f)	(g)	(h)	(i)	(j)	(k)	(l)	(m)	
(46)	Station Only	N/A	N/A	N/A	N/A	+1.80	+2.34	N/A	N/A	N/A	N/A	(46)
(47)	Regional (57 neighbors)	N/A	N/A	N/A	N/A	N/A	N/A	N/A	N/A	N/A	N/A	(47)

Nomenclature: See separate page